

CONFIDENTIAL -- CREDIT INTELLIGENCE REPORT

AI-Powered Credit Underwriting

Dashboard Section Reference Guide

A complete explanation of every underwriting parameter section, its data fields, AI model interactions, and risk flag interpretations.

Document Type:	Section Reference & AI Framework
Product:	Loan Against Property (LAP)
Audience:	Credit Officers - Risk Analysts - AI/ML Teams
Classification:	Internal -- Restricted Distribution

01 Identity	02 Employment	03 Income	04 Bureau	05 Property	06 Collateral
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How to Use This Report

Purpose

This document is the authoritative field-level reference for the AI-Powered Credit Underwriting Dashboard used in Loan Against Property (LAP) origination. Each section of the dashboard corresponds to a chapter here, explaining what the data means, how the AI model uses it, and what risk flags to watch for.

Dashboard Architecture

The dashboard is organised into six analytics cards rendered in a 3-column responsive grid. Each card corresponds to a distinct underwriting risk dimension. Cards are colour-coded by risk severity and contain animated gauges, progress bars, and KV metric grids designed for rapid visual triage by credit officers.

AI Integration

Every field described in this report is either a direct model feature or a source from which features are derived. Fields marked with risk flags are either hard gates (Boolean KO rules applied before ML scoring) or high-weight continuous features in the gradient boosted ensemble model.

NSTP System

Non-Standard Terms Processing (NSTP) is the deviation management framework. When any parameter falls outside policy norms, the AI system identifies the applicable NSTP rule code, determines the minimum authority level required, and routes the application to the appropriate approver queue automatically.

AI Framework Overview

HOW AI TRANSFORMS CREDIT UNDERWRITING

Traditional credit underwriting relied on rule-based scorecards, manual income verification, and subjective judgement. AI-powered underwriting replaces and augments each layer:

1. DATA INGESTION LAYER

Bureau data, bank statements, GST returns, property valuations, and KYC documents are parsed by OCR + NLP models. Structured features are extracted automatically, reducing processing time from days to minutes.

2. FEATURE ENGINEERING

Raw data is transformed into predictive signals: YoY revenue trend (velocity), DPD x recency interaction (severity-adjusted delinquency), enquiry velocity (credit-seeking behaviour), AMC/NMI ratio (income verification proxy), LTV x property tier (collateral risk), and 200+ additional derived features.

3. RISK SCORING MODEL

A gradient boosted ensemble (XGBoost + LightGBM) trained on 5+ years of LAP portfolio data produces a Composite Credit Score (0-100). Each of the six dashboard sections contributes a weighted sub-score. Approximate feature importance by section:

- Bureau & Credit (CIBIL): 28%
- Income & Financials: 22%
- Employment & Business: 18%
- Property Details: 16%
- Collateral & Legal: 9%
- Applicant Identity: 7%

4. DECISION ENGINE

Score bands map to decision outcomes:

80-100: Auto-Approve (no human intervention)

70-79: Approve via NCM NSTP (L_4 sign-off)

60-69: Approve via ZCM NSTP (L_3 sign-off)

50-59: Counter-offer (reduced loan amount/LTV)

0-49: Decline

5. NSTP ROUTING

Non-Standard Terms Processing triggers when individual parameter deviations are detected (e.g. CIBIL 684 -> NSTP rule APPLICANT_CIBIL_002). The AI system identifies the applicable NSTP authority level and auto-routes the file to the correct approver queue.

6. EXPLAINABILITY (XAI)

SHAP (SHapley Additive exPlanations) values are computed for every decision, providing regulators and credit officers with a human-readable explanation: "CIBIL score of 684 reduced approval probability by 18 points; business vintage of 13.98 years added 12 points."

7. CONTINUOUS LEARNING

The model is retrained quarterly on new originations and 12-month default outcomes. Portfolio drift is monitored via PSI (Population Stability Index) and CSI (Characteristic Stability Index) on all 200+ features.

Overview

The Applicant Identity section forms the foundational layer of the underwriting process. It captures the verified personal profile of the borrower -- who they are, where they live, and whether their identity documents are authentic and consistent.

Data Fields & Underwriting Significance

Age & Date of Birth	Determines eligibility windows, loan tenor limits (e.g. repayment before age 70), and regulatory age floors (typically 21-65 years).
Customer Segment	Classifies the applicant as SELF_EMPLOYED_PROFESSIONAL, SALARIED, or HUF. This drives income assessment method, multiplier tables, and NSTP routing.
Applicant Type	PRIMARY / CO-APPLICANT / GUARANTOR. Affects liability structure and credit weight assigned to each party's profile.
PAN & PAN NSDL Status	Primary tax identifier. NSDL real-time verification confirms the PAN is active, name-matched, and not flagged. A FALSE status is an automatic KO.
Aadhaar (masked)	Biometric-linked identity. Confirms residential address and enables de-duplication across bureau databases.
Nationality & Resident Type	NRI/RNOR applicants attract different LTV caps, income recognition rules, and repatriation compliance requirements.
Constitution	HUF, Proprietorship, Individual, or Partnership. Determines legal entity structure, signing authority, and applicable KYC norms.
Current Address	Geo-coded for branch/region mapping, property proximity check, and bureau address consistency validation.

How AI Uses This Section

AI models ingest identity signals as categorical features. Name-match fuzzy scoring across PAN/Aadhaar/bureau headers flags synthetic identities. Segment classification feeds the income model selector. Age at origination and age at maturity are continuous features in the eligibility neural network. NSDL status is a binary hard-gate -- a single FALSE halts the pipeline regardless of score.

Risk Flags & Triggers

PAN NSDL status = FALSE	-> automatic decline
Age at maturity > 70 years	-> tenor capped or guarantor required
Address mismatch between Aadhaar and bureau	-> manual KYC review
HUF constitution	-> additional Karta consent and coparcener NOC required

Overview

This section assesses the economic stability and income-generating capacity of the borrower's primary occupation. For Self-Employed Professionals (SEP) it analyses business maturity, financial audits, and turnover trends. For salaried applicants it evaluates employer quality and employment continuity.

Data Fields & Underwriting Significance

Business Vintage	Years of active operation. A minimum of 3 years is typically required; 5+ years signals established stability. The animated progress bar benchmarks against a 20-year sector median.
GST Turnover (Reported)	Revenue declared in GST returns. Used as the base for income computation under the ITR + GST method for SEP applicants.
Turnover Trend (YoY)	Year-over-year change in top-line revenue. A declining trend (v) triggers cashflow stress analysis and may require additional collateral or lower LTV.
TGE (Top Gross Entity)	Flags whether the entity being assessed is the primary income source. Prevents income double-counting across group structures.
Financials Audited	Confirms CA-certified books. Unaudited financials may be discounted by 20-30% in income computation.
Employer / Entity Name	For salaried: employer quality tier (Public Sector, MNC, Listed, Unlisted Private). For SEP: legal entity name cross-referenced with GST and MCA filings.
Total Work Experience	Continuous employment history. Gaps > 3 months in the last 2 years flag income instability risk.
Income Multiplier	Applied to Net Monthly Income to compute maximum loan eligibility. SEP category attracts 25x; Salaried may receive up to 36x depending on employer tier.

How AI Uses This Section

The AI income engine branches on the Segment feature: SEP applicants are routed to the turnover-based model; Salaried to the payslip model. The YoY trend is a gradient feature -- negative slope exceeding 15% triggers a stress-test on repayment capacity under revenue contraction scenarios. Business vintage is log-transformed and combined with turnover volatility to produce a 'Business Stability Score' (0-100). Audit certification status is a binary multiplier on computed income.

Risk Flags & Triggers

Turnover decline > 25% YoY	-> income haircut applied, LTV reduced
Business vintage < 3 years	-> sub-prime bracket, NSTP required
Unaudited financials	-> income discounted 25%
TGE flag absent	-> income verification escalated to credit ops

Overview

The Income & Financials section quantifies the borrower's repayment capacity -- the single most critical variable in credit decisioning. It combines verified income, debt obligations, and banking behaviour into a holistic affordability picture.

Data Fields & Underwriting Significance

Net Monthly Income (NMI)	Eligible income after professional tax, existing EMI obligations, and any haircuts. The foundational input for EMI affordability computation.
Total Eligible Income (Annual)	Annualised NMI. Used for tax bracket verification and macro-level sanity checks against declared ITR.
Income Program Applied	The underwriting program (e.g. CROSS_SELL_BT, STANDARD_LAP). Each program has distinct income recognition rules, LTV caps, and NSTP thresholds.
DBR / DTI Ratio	Debt Burden Ratio -- total monthly EMI obligations as a percentage of NMI. Policy threshold is 50%; above 40% triggers review. The circular gauge provides instant visual risk classification.
Total Monthly Obligation	Sum of all active EMIs including proposed EMI. An elevated obligation relative to income is the most common trigger for application decline.
Avg Monthly Credits (AMC)	Average monthly inflows across all bank accounts over 6-12 months. Serves as an independent income proxy to cross-validate declared NMI.
AMC vs Proposed EMI	Coverage ratio = AMC ÷ Proposed EMI. A ratio >= 3.0x signals strong cashflow buffer. Below 1.5x indicates cashflow stress.
Cheque Returns / Bounces	Count of dishonoured instruments in the last 6 months. A proxy for liquidity stress and payment discipline. More than 3 triggers mandatory review.

How AI Uses This Section

DBR is the primary continuous feature in the repayment capacity model. The AI model combines NMI, AMC, obligation stack, and cheque bounce frequency into a multi-variate 'Cashflow Health Score'. The AMC/EMI coverage ratio feeds a time-series model trained on historical default rates segmented by income program. Bounce count is a non-linear risk amplifier -- the marginal default probability per additional bounce increases exponentially beyond 2 events. The DBR gauge colour-codes risk at four thresholds: <30% Healthy, 30-40% Moderate, 40-50% Elevated, >50% High Risk.

Risk Flags & Triggers

DBR > 50%	-> automatic decline in most programs
AMC/EMI coverage < 1.5x	-> escalation to credit manager
Bounces L6M >= 3	-> NSTP required; >= 5 -> decline
NMI/ITR variance > 20%	-> income verification flag

Overview

The Bureau & Credit section is the external credit intelligence layer. It translates the borrower's entire credit history -- spanning all lenders -- into standardised risk signals. CIBIL score, delinquency depth, overdue balances, and enquiry patterns collectively determine creditworthiness and NSTP authority level.

Data Fields & Underwriting Significance

CIBIL Score	Ranges 300-900. The animated semicircular gauge positions the score within Poor/Fair/Excellent bands. Policy minimum is 750; 650-749 triggers NSTP; below 650 is a decline KO in most programs.
CMR Rank	Company/Credit Management Rank for non-individual entities (HUF, firms). Ranks 1-10 where 1 is best. Rank ≥ 7 triggers review; used alongside CIBIL score for SEP applicants.
Max DPD (Last 12M)	Maximum Days Past Due across all live accounts in the past 12 months. Policy requires 0 DPD; any DPD > 0 is a deviation. DPD > 30 is typically a hard KO.
Enquiry Count (L6M)	Number of credit bureau enquiries in the last 6 months. High enquiry count signals credit-seeking behaviour and potential overleveraging. >5 triggers risk review.
Overdue Amount CC/KCC	Outstanding overdue on credit card and Kisan Credit Card accounts. Must be Rs.0 at disbursement. Any overdue triggers a pre-disbursement clearance condition.
Overdue Amount Non-CC	Overdue on non-revolving credit facilities. Same zero-tolerance policy applies.
Derogatory Status	Indicates written-off, settled, or willful default accounts. A non-zero derogatory code is an immediate decline trigger in virtually all programs.
Suit Filed / Wilful Default	Legal proceedings or RBI willful defaulter list inclusion. Hard KO -- no exception or NSTP authority can override this flag.
Days Since Last Payment	Recency of credit activity. Very old last payment (>3 years) may indicate dormant credit profile, complicating score interpretation.

How AI Uses This Section

CIBIL score is the single most predictive feature in the default model -- contributing approximately 28% of feature importance in gradient boosted tree models. The AI system uses the score band (not raw score) as a categorical routing variable: ≥ 750 routes to standard processing; 650-749 to NSTP L_3/L_4; <650 triggers pre-decline review. DPD is modelled as a severity x recency interaction term. The enquiry velocity (enquiries per month trend) is a derived feature that detects credit desperation patterns not visible in the raw count. Derogatory and suit flags are hard Boolean gates applied before the ML score is computed -- they short-circuit the entire pipeline.

	Risk Flags & Triggers	
	CIBIL < 650	-> decline in standard program; review in special programs
	CIBIL 650-749	-> NSTP required (L_3 NCM or higher)
	Any DPD > 0 in L12M	-> deviation; DPD > 30 -> decline
	Derogatory code != 0	-> hard KO
	Suit Filed / Wilful Default	-> hard KO, no override
	Overdue > Rs.0	-> pre-disbursement clearance condition

Overview

The Property Details section evaluates the collateral underpinning the Loan Against Property. In LAP underwriting, the property is both the source of security and a key risk variable -- its type, location tier, market value, and LTV determine the maximum sanctionable loan amount and influence the risk premium.

Data Fields & Underwriting Significance

Property Type	Residential, Commercial, or Industrial. Each type has distinct LTV caps (Residential: up to 70%; Commercial: up to 60%; Industrial: up to 55%) and liquidity risk profiles.
Property Sub-Type	Further classifies into Self-Occupied, Rented-out, Vacant, etc. A rented investment property introduces tenancy risk and affects forced-sale value assumptions.
Tier Location	Tier 1 (metros), Tier 2, or Tier 3 cities. Location tier directly impacts liquidity discount applied to market value. Tier 3 properties may attract 20-30% additional haircut.
Min Market Value	Lower bound of the bank-empanelled valuer's assessment. Loan computation uses the minimum -- not average -- to build in a conservative buffer.
Loan Amount	Proposed disbursement amount. Must satisfy: $\text{Loan Amount} \leq \text{Market Value} \times \text{LTV Cap}$.
LTV Ratio (Policy)	Loan-to-Value ratio. The animated donut gauge provides immediate visual classification: <50% Strong (green), 50-65% Moderate (amber), >65% High Risk (red). RBI caps LAP LTV at 70%.
Max Loan by LTV	Computed ceiling: $\text{Market Value} \times \text{Policy LTV\%}$. The proposed loan must not exceed this figure. Breaches require RBI exception or valuer re-assessment.

How AI Uses This Section

The property risk model is a geo-spatial ML model trained on historical property liquidation data, title dispute rates, and forced-sale-value/market-value ratios across pin codes. Location tier is encoded using target-encoded district-level default rates. LTV ratio is the strongest collateral feature -- a 10% increase in LTV above 60% is associated with a 34% increase in loss-given-default in historical LAP portfolios. Property sub-type interacts with location tier: a Tier 2 commercial property has significantly lower liquidity than a Tier 1 residential property at the same LTV. The AI model computes a 'Collateral Adequacy Score' (0-100) combining all property signals.

Risk Flags & Triggers		
LTV > 65%	->	amber risk; LTV > 70% -> RBI ceiling breach
Tier 3 location	->	additional 15% value haircut in model
Commercial/Industrial type	->	lower LTV cap applied
Rented-out property	->	tenancy risk flag, legal opinion mandatory
Market value < Rs.30L	->	minimum loan size constraint may apply

Overview

The Collateral & Legal section consolidates fraud detection signals, legal due diligence status, and operational processing metrics. It is the final compliance gate before sanction -- ensuring the property title is clear, the applicant has no fraud ring linkages, and all legal instruments are in order.

Data Fields & Underwriting Significance

Hunter / Fraud Ring Detection	KARZA API real-time check against known fraud ring databases. A 'Match' result is a hard KO. 'No Match' (isMatch=FALSE) is required to proceed.
External Dedupe	Cross-checks the applicant against all active and historical loan applications within the lending institution and partner network. Prevents concurrent application fraud.
Legal Report Processing Days	Calendar days elapsed since the legal search report was commissioned to the panel advocate. Benchmark is <90 days; >200 days signals systemic delay and may require re-commissioning.
Technical Report Processing Days	Days elapsed for the panel valuer's technical/valuation report. >300 days indicates a stale valuation that may not reflect current market conditions.
EMI Paid Count	Number of EMIs paid on record across existing loan accounts. A high count (>24) signals consistent payment history; used as a positive signal in the repayment model.
Bounce Count (L6M)	As in the Income section, ECS/NACH bounce count on existing loan EMIs. A direct measure of payment discipline on current obligations.
Title Clearance	Certificate confirming the property has a clean, marketable title -- no encumbrances, disputes, or litigation. Mandatory for disbursement.
Encumbrance Certificate	Confirms the property has no registered charges, mortgages, or legal liabilities for the requested search period. Required from the sub-registrar's office.

How AI Uses This Section

Fraud detection is a rule-based pre-filter applied before ML scoring. Hunter match and dedupe flags are Boolean hard gates. The AI system then applies a 'Legal Risk Score' computed from report processing delays (used as proxies for document quality issues), title clearance status, and encumbrance certificate recency. Processing delays > 300 days are treated as a missing data signal -- the model assigns higher uncertainty to the collateral valuation and may recommend a conditional sanction pending fresh reports. EMI paid count and bounce count from existing loans are recycled as payment behaviour features, reinforcing signals already captured in the Bureau section but sourced from internal loan management systems rather than the external bureau.

Risk Flags & Triggers		
Hunter match = TRUE	->	hard KO, fraud alert raised
External dedupe match	->	investigation required before processing
Legal report > 300 days	->	re-commission required
Technical report > 300 days	->	fresh valuation mandatory
Title clearance pending	->	conditional sanction only
Encumbrance cert pending	->	disbursement blocked

Quick Reference Risk Flag Matrix

Section	Parameter	Threshold	Action
Identity	PAN NSDL Status	= FALSE	Hard KO -- Decline
Identity	Age at Maturity	> 70 years	Tenor cap / Guarantor
Employment	Business Vintage	< 3 years	NSTP required
Employment	Turnover Trend YoY	Decline > 25%	Income haircut + LTV cut
Employment	Financials Audited	= No	Income discounted 25%
Income	DBR / DTI	> 50%	Hard KO -- Decline
Income	DBR / DTI	40-50%	NSTP required
Income	AMC / EMI Coverage	< 1.5x	Escalate to CM
Income	Bounces L6M	>= 3	NSTP; >= 5 -> Decline
Bureau	CIBIL Score	< 650	Hard KO -- Decline
Bureau	CIBIL Score	650-749	NSTP L_3/L_4 required
Bureau	Max DPD L12M	> 30 days	Hard KO -- Decline
Bureau	Derogatory Code	!= 0	Hard KO -- Decline
Bureau	Suit Filed	= Yes	Hard KO -- No override
Bureau	Overdue Amount	> Rs.0	Pre-disbursement clear
Property	LTV Ratio	> 70%	RBI ceiling breach
Property	LTV Ratio	65-70%	High risk review
Property	Location Tier	= Tier 3	Haircut 15% on value
Collateral	Hunter Match	= TRUE	Hard KO -- Fraud alert
Collateral	Dedupe Match	= Match	Investigation required
Collateral	Legal Report Age	> 300 days	Re-commission required
Collateral	Title Clearance	= Pending	Conditional sanction
Collateral	Encumbrance Cert	= Pending	Disbursement blocked